

## Chapter 31 — Tuberculosis and Parasitic Infections of the Genitourinary Tract — Numbers, Gold Standards and Traps

### IF YOU READ NOTHING ELSE

- **Urogenital tuberculosis (UG TB) gold standard:** urine acid-fast bacilli (AFB) culture, first-void, 3–5 consecutive days.
- **Hallmark:** sterile pyuria ± haematuria.
- **Kidney = 80% of UG TB;** epididymis second haematogenous site.
- **Standard regimen:** 2 months isoniazid (INH) + rifampicin + pyrazinamide (+ ethambutol) → 4 months INH + rifampicin.
- **Surgery timing:** 4–6 weeks after starting anti-TB drugs.
- **UG TB relapse** 6.3–22% → follow up to 10 years.
- **Schistosomiasis gold standard:** eggs in urine (collect 09:00–15:00, peak noon).
- **Praziquantel (PZQ)** 40 mg/kg (2 × 20 mg/kg, 6–8 h apart) = only WHO drug.
- **Schistosomal bladder cancer:** squamous cell carcinoma (SCC) 60–90%, age 40–50.
- **Lymphatic filariasis (LF):** diethylcarbamazine (DEC) = drug of choice; never DEC in onchocerciasis.

### UROGENITAL TUBERCULOSIS

- **Commonest:** *M. tuberculosis* then *M. bovis*; *M. bovis* innately pyrazinamide-resistant.
- **UG TB:** 15–20% of pulmonary TB (developing) vs 2–10% (developed).
- **Extrapulmonary rank:** 2nd after nodes (developing); 3rd after pleural, lymphatic (US).
- **Male : female** about 2 : 1; adults.
- **Latency:** 90–95% controlled; <5% primary progression; reactivation 5–15% lifetime.
- **Latency to UG disease** up to 46 years.
- **Route:** haematogenous (principal) → kidney, epididymis.
- **Intravesical bacille Calmette-Guérin (BCG)** → UG TB in about 0.9%.
- **UG TB not infectious** unless aerosol-generating procedure or skin inoculation.
- **Constitutional symptoms** <20%; renal colic <10%.
- **Autonephrectomy up to 33%;** end-stage renal disease about 7%.
- **Ureteric stricture** commonest at vesicoureteric junction; pan-ureteric → beaded corkscrew.
- **Bladder:** dome first, trigone spared; symptoms at <100 mL; thimble <20 mL.
- **Epididymis** 10–55%, bilateral one-third; starts in globus minor; scrotal sinus up to 50%.
- **Prostate** 22–49%; urethra 1.9–4.5%; hydrocele about 5%.
- **Coexisting pulmonary TB** up to 20% → chest radiograph for all.

### Diagnosis

- **Culture:** Löwenstein-Jensen 4–6 weeks; Mycobacteria Growth Indicator Tube (MGIT) ≥10 days; always + solid medium.
- **Ziehl-Neelsen urine** sensitivity <50%.
- **GeneXpert:** TB + rifampicin resistance in 1–2 days; rifampicin resistance = multidrug-resistant (MDR) surrogate.
- **Nucleic acid amplification test (NAAT):** adjunct only; never excludes TB; useless for monitoring.
- **Lateral-flow urine lipoarabinomannan (LF-LAM):** HIV + symptoms, advanced HIV, or CD4 <200.
- **Tuberculin skin test (TST) cut-offs:** ≥5 mm immunosuppressed or contact; ≥10 mm at-risk; ≥15 mm no risk.
- **Interferon-γ release assay (IGRA)** preferred if ≥5 years, born in endemic country.
- **T-SPOT.TB** most sensitive IGRA; IGRA unaffected by BCG.
- **Bladder biopsy** 19–52% sensitive if culture-negative.

### Imaging eponyms and signs

- **Kerr's kink:** hiked-up pelvis, sharp pelviureteric junction angulation.

- **Lobar calcification ("putty"/"cement" kidney)** = pathognomonic end-stage TB.
- **Phantom calyx:** stenosed calyx fails to opacify.
- **Moth-eaten calyx** = early intravenous urography (IVU) sign.
- **Pipestem or beaded-corkscrew ureter** → TB.
- **Golf-hole ureteric orifice** → image upper tract.
- **Computed tomography (CT) urography** = commonest modality (developed world).
- **Ultrasound** → monitor hydronephrosis on treatment.

### Treatment

- **Toxicities:** INH → neuropathy (give pyridoxine); ethambutol → red-green vision; pyrazinamide → urate, arthralgia.
- **All first-line except ethambutol** hepatotoxic.
- **Stop ethambutol** once fully susceptible.
- **Rifabutin** → uveitis; rifamycins → antiretroviral and immunosuppressant interactions.
- **Contraindicated:** rifapentine in HIV, cavitory or extrapulmonary TB.
- **No twice-weekly continuation;** ≥3×/week only.
- **Complicated UG TB:** 12–24 months, or ≥6 months after surgery.
- **Steroids:** proven only in TB meningitis; UG use anecdotal.
- **Surgery needed** in about 55%; reconstruction >half of operations.
- **Obstruction:** retrograde double-J first (41% success) → antegrade stent → nephrostomy (PCN).
- **Nephrectomy if nonfunctioning kidney** + recalcitrant TB or resistant hypertension (improves 65%).
- **Open surgery** if no improvement after 6 weeks of drugs.
- **Gaps:** bladder mobilisation 2–3 cm; psoas hitch ≤5 cm; Boari 10–15 cm.
- **Augmentation** <100 mL; **substitution** for thimble <20 mL.
- **Emergency drainage** if uraemia or sepsis; avoid high contrast pressure.
- **Epididymectomy** if testis spared; vas ligation at external ring.
- **Follow-up:** pulmonary 2 years vs UG up to 10 years (mean relapse 5.3 years).

### Special situations

- **MDR:** INH + rifampicin; cure 50–60% vs 94–97%.
- **Pre-XDR** (pre-extensively drug-resistant) = MDR + fluoroquinolone OR injectable.
- **XDR:** MDR + fluoroquinolone + injectable, or bedaquiline/linezolid; cure 30–50%.
- **BPAL** (bedaquiline, pretomanid, linezolid) 6–9 months; not approved extrapulmonary.
- **Pregnancy:** INH + rifampicin + ethambutol + pyridoxine × 9 months; breastfeeding allowed.

- **HIV:** 15–20× risk; 6 months daily; 9 months if not on antiretrovirals.
- **Transplant:** within 6 months; fever; rifamycin-free ≥18 months; mortality up to 36%.

## PARASITIC INFECTIONS OF THE UROGENITAL TRACT

### Schistosomiasis

- ***S. haematobium*:** vesical plexus, eggs in urine, *Bulinus* snail.
- **Egg:** about 140 µm, **terminal spine**.
- **Bilharz**, Cairo 1852: worms linked to eggs.
- **Adults** live 3–5 years on average; cercaria → adult 80–110 days.
- **No cercarial shedding** <15 °C or >35 °C.
- **Peak burden** age 5–15; dams increase transmission (Aswan).
- **Obstruction commonest** at juxtavesical ureter.
- **Hydroureter precedes hydronephrosis**; tubular function lost before glomerular.
- **Tonic hydroureter** → reversible hydronephrosis; segmental rarely significant; atonic aperistaltic.
- **International Agency for Research on Cancer (IARC) Group 1** carcinogen.
- **Mass drug administration (MDA), Egypt:** bladder cancer 28% → 12%, shift SCC → urothelial.
- **Katayama fever:** 2–8 weeks; mainly *S. japonicum*; steroids + PZQ.
- **Oviposition** from 8–12 weeks; serology positive 4–6 weeks.
- **Contracted bladder** ≤50 mL.
- **End stage:** 40–60% of obstructive uropathy presents here.
- **Female genital schistosomiasis** in 33–75% of infected women.
- **Male genital schistosomiasis:** diagnose by eggs in ejaculate.
- **Vesicoureteric reflux** in 25% of infected ureters.
- **PZQ efficacy** 60–90%; obstruction resolves in 1–2 months; polyps in children 2–6 weeks.
- **Rectal snip before bladder biopsy**; squash preparation shows viability.
- **Circulating antigens (CAA/CCA)** = active infection only; serology stays positive after cure.
- **Artemisinins** = prophylaxis (schistosomulae); not in malaria-coendemic MDA.
- **Deep chronic ulcer** → **partial cystectomy** (not fulguration).

- **Mortality:** obstructive uropathy in 20s; cancer and pyelonephritis >40.
- **Egg input must fall about 90%** to cut transmission.
- **WHA65.21 (2012):** shift from morbidity control to elimination.

### Filariasis

- ***W. bancrofti*** >90% of LF; only species with male genital disease.
- **Microfilaraemia peaks at midnight**; Giemsa smear.
- **Adults** 5–7 years; mature over 6–9 months.
- **Bancroftian lymphoedema** whole limb vs *Brugia* below knee.
- **Acute adenolymphangitis (ADL)** retrograde vs dermatolymphangioadenitis (DLA) proximal and bacterial.
- **Filarial dance sign** on ultrasound: ≥80% sensitive.
- **Antigen test** for *W. bancrofti* only; none for *Brugia*.
- **DEC** 2 mg/kg three times daily; 1 day = 12 days.
- **Doxycycline** 200 mg daily for 4–8 weeks → sterilises adults (*Wolbachia*).
- **Filarial hydrocele pointers:** cord/epididymal nodules, thick tunica with cholesterol or calcium.
- **Hydrocelectomy:** excise intact sac; ligate leaking lymphatics.
- **Tropical pulmonary eosinophilia:** nocturnal cough, high IgE; microfilaraemia absent; DEC 2–3 weeks.
- **Contraindicated:** doxycycline in pregnancy and young children.
- **MDA:** albendazole + ivermectin (Africa), albendazole + DEC (elsewhere).
- **Onchocerciasis:** *Simulium* fly; hanging groin; ivermectin then doxycycline.
- ***Loa loa*:** *Chrysops* fly; no *Wolbachia*; microfilariae >2500–8000/mL → encephalopathy risk.

### Other parasites

- **Hydatid kidney** about 2–3%; PAIR (puncture, aspiration, injection, reaspiration) not for renal cysts; albendazole 1–6 months.
- **Hydatid cyst rupture** → anaphylaxis.
- **Pinworm:** single albendazole or mebendazole; treat household; repeat at 2 weeks.
- **Amoebiasis:** tinidazole or metronidazole → then luminal agent (paromomycin or iodoquinol).

### CLASSIC TRAPS

- **Trap:** sterile pyuria = partially treated urinary tract infection — **Actually:** think UG TB.
- **Trap:** negative TST/IGRA excludes active TB — **Actually:** neither rules TB in or out.
- **Trap:** negative NAAT excludes UG TB — **Actually:** adjunct only; culture remains standard.
- **Trap:** treatment always improves obstruction — **Actually:** healing fibrosis can worsen it; keep imaging.
- **Trap:** 4-month rifapentine–moxifloxacin regimen for UG TB — **Actually:** not for extrapulmonary TB.
- **Trap:** ureteric calcification same in TB and bilharzia — **Actually:** TB intraluminal, nondilated; bilharzia intramural, dilated.
- **Trap:** PZQ aborts early infection — **Actually:** poor against schistosomulae; repeat after weeks.
- **Trap:** bladder biopsy first for schistosomiasis — **Actually:** rectal snip first; squash preparation.
- **Trap:** treat *Salmonella* bacteriuria with antibiotics alone — **Actually:** add PZQ; bacteria shelter in tegument.
- **Trap:** DEC for all filariae — **Actually:** never in onchocerciasis; caution with *Loa loa*.